

The Tone Field Manual

Dial in a great sound on any amp in five minutes

A routine that works on unfamiliar amps, what every knob is actually doing, the guitar EQ map in players' words, and a troubleshooting table for the tones everyone fights.

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Free from guitarhub.org. Share it, print it, pin it above the amp. The interactive versions of these ideas live at guitarhub.org/tone.

The five-minute dial-in

Use this on any amp, including the backline at a venue. It works because it changes one thing at a time and checks each change against the one before.

- 1 Reset.** Every EQ knob to noon (5), gain low, master or volume at a sensible level for the room. Guitar volume and tone on 10.
- 2 Set gain by playing.** Play the hardest part you will play tonight. Raise gain until it has the grit you want, then back it off a little. Most people use more gain than the song needs.
- 3 Mids first.** Sweep the mid knob fully both ways and settle where chords sound most like a guitar and least like a radio. Mids decide whether you will be heard.
- 4 Bass until tight.** Raise bass until palm mutes feel solid; stop before they boom or the low strings smear.
- 5 Treble until clear.** Raise treble until the pick attack is defined; stop before single high notes sting.
- 6 Presence and reverb last.** Small moves. If the amp sounds great alone with lots of reverb, halve it for the band.
- 7 Check at the real volume.** Ears hear more bass and treble when loud. Re-check at performance volume and trim.

The one rule: change one knob, play the same phrase, compare. Two changes at once teach you nothing.

Ten-second gain check

Play a big open chord and let it ring. If you can still hear each string, the gain is fine for chords. If it turns into one blurry roar, back off.

What every amp knob does

Knob	What it really controls	Start here
Gain / Drive	How hard the preamp is pushed, so how much it distorts. Also adds compression and noise.	Low, then raise by ear.
Master / Volume	Loudness after the preamp. On single-volume amps it also sets distortion.	As loud as the room allows.
Bass	Low end and weight. On many amps it interacts with mid and treble.	Noon, then lower for high gain.
Middle	Body and presence in a mix. The most important band in a band.	Noon or above.
Treble	Clarity and pick attack in the preamp.	Noon, adjust by ear.
Presence	Upper treble in the power section. Adds air or glass.	Below noon; raise slowly.
Reverb	How far back in a room you sound.	Less than feels right alone.
Bright switch	Treble boost that fades as volume rises.	On for dark guitars at low volume.

Interactive tone stacks. On many classic designs the three EQ knobs share one circuit, so turning mids down also changes how bass and treble behave, and all three at zero can mean almost silence. Treat them as a set, not three separate controls.

Guitar controls are amp controls

Rolling guitar volume from 10 to 7 cleans up a driven amp without touching it. The tone knob around 6 or 7 tames harsh bridge pickups. Learn to ride both while playing.

The guitar EQ map

Every region has a word players use when there is too much of it. Learn the words and any tone becomes fixable.

Range	Word	Too much	Too little	Notes
80 to 120 Hz	Boom	Flubby, fights bass and kick	Thin, no weight alone	High-pass around 80 to 100 Hz in a band mix
200 to 300 Hz	Mud	Cloudy, chords blur	Cold, papery	Cut here before adding treble
400 to 600 Hz	Boxy	Hollow, cardboard	Scooped, distant	A small cut often cleans up cheap cabs
700 to 1,000 Hz	Honk	Nasal, horn-like	Lacks presence	Where wah and cocked-wah tones live
1.2 to 2 kHz	Push	Pokey, in your face	Buried in the band	The cut-through band. Rarely scoop it.
2.5 to 4 kHz	Bite	Harsh, ear fatigue	Dull, no attack	Find the harsh spot and cut 2 to 3 dB
5 kHz and up	Fizz	Buzzy distortion, hiss	Dark, blanketed	Speakers and cab IRs roll most of this off

Cut before you boost. Boosting adds level and gain and changes everything downstream. Cutting removes only the problem.

Why a scooped tone disappears

Lots of bass and treble with the mids cut sounds huge alone. In a band the bass owns the lows and the cymbals own the highs, so a scooped guitar has nothing left. Put the mids back and it reappears.

Train these by ear with the free EQ Ear Trainer at guitarhub.org/eq-ear-trainer.

Tone troubleshooting

Problem	Usual causes	Try first
Thin	Bass too low, bridge pickup with tone at 10, pickups too far from strings	Bass to noon, try the middle or neck pickup, raise pickup height a little
Muddy	Too much low-mid, too much gain, neck pickup into high gain	Cut 200 to 300 Hz, lower gain, trim bass before the gain stage
Harsh	Treble or presence high, bright pickups, no speaker simulation	Guitar tone knob to 7, cut 3 kHz, check the cab or IR is on
Buried in the band	Mids scooped, too much reverb, too much gain	Mids up, reverb mix down, less gain than you use alone
Fizzy distortion	Gain too high, no cab sim, pedal tone too bright	Gain down, cab on, pedal tone below noon
No sustain	Old strings, low gain, dead notes from setup	New strings, a compressor or a little more gain, check the setup
Lifeless, squashed	Too much compression and gain stacked	Remove one gain stage; compressor sustain down

Hands and guitar checklist

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| ☐ Strings less than a few months old | ☐ Tried picking over the neck and near the bridge |
| ☐ Intonation checked at the 12th fret | ☐ Tried a thicker and a thinner pick |
| ☐ Pickup heights balanced neck to bridge | ☐ Guitar volume used as a gain control |
| ☐ Action comfortable, no fret buzz | ☐ Tone knob used, not left on 10 |